

Mathematical Background for the SVD–LDA Project

Numerical Methods and Algorithms

1 Linear algebra preliminaries

We work over the real numbers. Vectors are columns, and matrices are written with capital letters. For a vector $x \in \mathbb{R}^n$, we use the Euclidean norm

$$\|x\|_2 = \sqrt{x^\top x}.$$

For a matrix $A \in \mathbb{R}^{m \times n}$, the Frobenius norm is

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2} = \sqrt{\text{trace}(A^\top A)}.$$

A square matrix $A \in \mathbb{R}^{n \times n}$ is *symmetric* if $A = A^\top$. A real symmetric matrix has the following important properties:

- All eigenvalues are real.
- There exists an orthonormal basis of eigenvectors; i.e. $A = Q\Lambda Q^\top$ with Q orthogonal and Λ diagonal.

An orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ satisfies $Q^\top Q = I_n$. Its columns are an orthonormal basis of $\mathbb{R}^n$.

2 Singular value decomposition (SVD)

2.1 Definition and basic properties

Theorem 1 (Singular value decomposition). *Let $A \in \mathbb{R}^{m \times n}$ have rank r . Then there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ and a diagonal matrix*

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & & & & 0 & \\ & & & & & \ddots \end{bmatrix} \in \mathbb{R}^{m \times n}$$

with singular values

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

such that

$$A = U\Sigma V^\top.$$

The columns of U are called *left singular vectors* of A ; the columns of V are *right singular vectors*.

Proposition 1 (Relation to eigenvalues). *Let $A = U\Sigma V^\top$ be an SVD. Then*

$$A^\top A = V\Sigma^\top \Sigma V^\top, \quad AA^\top = U\Sigma\Sigma^\top U^\top.$$

In particular,

- *The eigenvalues of $A^\top A$ and $AA^\top$ are $\sigma_1^2, \dots, \sigma_r^2$ (and possibly some zeros).*
- *The right singular vectors of A are eigenvectors of $A^\top A$.*
- *The left singular vectors of A are eigenvectors of $AA^\top$.*

Thus one can think of the singular values as square roots of the eigenvalues of $A^\top A$.

2.2 Truncated SVD and best rank- k approximation

Let $A = U\Sigma V^\top$ be an SVD with singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. We can write A as

$$A = \sum_{i=1}^r \sigma_i u_i v_i^\top,$$

where u_i and v_i are the i -th columns of U and V , respectively.

For an integer k with $1 \leq k \leq r$, the *truncated SVD of rank k* is

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^\top.$$

Theorem 2 (Best rank- k approximation (Eckart–Young)). *Let $A \in \mathbb{R}^{m \times n}$ have SVD $A = U\Sigma V^\top$. For $1 \leq k \leq r$, the truncated SVD A_k solves*

$$\|A - A_k\|_F = \min_{\text{rank}(B) \leq k} \|A - B\|_F.$$

Moreover,

$$\|A - A_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2.$$

Thus among *all* matrices of rank at most k , A_k is the closest to A in Frobenius norm.

2.3 Energy and effective rank

Define the “energy” (squared Frobenius norm) of A by

$$\|A\|_F^2 = \sum_{i=1}^r \sigma_i^2.$$

The energy captured by the first k singular values is

$$E(k) = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{j=1}^r \sigma_j^2}.$$

By the theorem above, $\|A - A_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2$, so $E(k)$ measures how much of the total variance (or information) is preserved by A_k .

Definition 1 (Effective rank). Fix a threshold $\alpha \in (0, 1)$, for example $\alpha = 0.9$ or $\alpha = 0.95$. The effective rank of A at level α is

$$r_\alpha(A) = \min\{k : E(k) \geq \alpha\}.$$

If $r_{0.9}(A)$ is small, then most of the information in A can be captured in a low-dimensional subspace.

3 The power method for eigenvalues

Suppose $A \in \mathbb{R}^{n \times n}$ is real symmetric, with eigenvalues

$$\lambda_1, \lambda_2, \dots, \lambda_n,$$

ordered so that $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$. Let $q_1, \dots, q_n$ be an orthonormal basis of eigenvectors, so $Aq_i = \lambda_i q_i$.

3.1 Algorithm

The *power method* approximates the dominant eigenvalue λ_1 and its eigenvector.

Given a starting vector $x^{(0)} \neq 0$ and integers $k = 0, 1, 2, \dots$, define

$$y^{(k+1)} = Ax^{(k)}, \quad x^{(k+1)} = \frac{y^{(k+1)}}{\|y^{(k+1)}\|_2}.$$

At iteration k we can form the Rayleigh quotient

$$\lambda^{(k)} = (x^{(k)})^\top Ax^{(k)}.$$

We stop when $|\lambda^{(k)} - \lambda^{(k-1)}|/|\lambda^{(k)}|$ is below a tolerance.

3.2 Convergence idea

Write $x^{(0)}$ in the eigenbasis:

$$x^{(0)} = c_1 q_1 + c_2 q_2 + \dots + c_n q_n,$$

with $c_1 \neq 0$ (this holds for almost all starting vectors).

Then

$$A^k x^{(0)} = c_1 \lambda_1^k q_1 + c_2 \lambda_2^k q_2 + \dots + c_n \lambda_n^k q_n = \lambda_1^k \left(c_1 q_1 + c_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k q_2 + \dots \right).$$

Since $|\lambda_2/\lambda_1| < 1$, the terms with $i \geq 2$ decay geometrically. After normalization, $x^{(k)}$ converges to $\pm q_1$, and $\lambda^{(k)}$ converges to λ_1 .

The rate of convergence is governed by the ratio $|\lambda_2/\lambda_1|$; if the dominant eigenvalue is well separated, the method converges faster.

4 PCA and its connection to SVD

Principal Component Analysis (PCA) is a method for reducing the dimension of data while preserving as much variance as possible.

4.1 Data matrix and covariance

Suppose we have N data vectors $x_1, \dots, x_N \in \mathbb{R}^d$. Stack them into a data matrix

$$X = \begin{bmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_N^\top \end{bmatrix} \in \mathbb{R}^{N \times d}.$$

Let $\mu \in \mathbb{R}^d$ be the sample mean

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i.$$

We define centered data $\tilde{x}_i = x_i - \mu$ and the centered data matrix

$$\tilde{X} = \begin{bmatrix} \tilde{x}_1^\top \\ \vdots \\ \tilde{x}_N^\top \end{bmatrix}.$$

The sample covariance matrix is

$$C = \frac{1}{N-1} \tilde{X}^\top \tilde{X} \in \mathbb{R}^{d \times d}.$$

4.2 PCA as an eigenvalue problem

The principal components are defined as eigenvectors of C . If

$$Cv_j = \lambda_j v_j,$$

with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$, then v_1 is the direction along which the projected data has maximum variance; v_2 is the direction of next-largest variance, and so on.

Projecting onto the first k principal components gives a k -dimensional representation

$$z_i = \begin{bmatrix} v_1^\top \tilde{x}_i \\ \vdots \\ v_k^\top \tilde{x}_i \end{bmatrix} \in \mathbb{R}^k.$$

4.3 Connection to SVD

Consider the SVD of the centered data matrix:

$$\tilde{X} = U \Sigma V^\top, \quad \tilde{X} \in \mathbb{R}^{N \times d}.$$

Then

$$\tilde{X}^\top \tilde{X} = V \Sigma^\top \Sigma V^\top.$$

Thus

- The columns of V are eigenvectors of C .
- The eigenvalues of C are proportional to the squared singular values:

$$C = \frac{1}{N-1} \tilde{X}^\top \tilde{X} = V \left(\frac{\Sigma^\top \Sigma}{N-1} \right) V^\top.$$

In particular, PCA can be computed via the SVD of the centered data matrix.

5 Two-class Linear Discriminant Analysis (LDA)

Whereas PCA is unsupervised (it does not use labels), Linear Discriminant Analysis (LDA) is a *supervised* method that aims to find directions that separate classes.

We focus on the two-class case with labels 0 and 1.

5.1 Scatter matrices

Let X_0 be the set of feature vectors from class 0 and X_1 from class 1. Let

$$\mu_0 = \frac{1}{n_0} \sum_{x \in X_0} x, \quad \mu_1 = \frac{1}{n_1} \sum_{x \in X_1} x$$

be class means and μ the overall mean. The *within-class scatter matrix* is

$$S_W = \sum_{x \in X_0} (x - \mu_0)(x - \mu_0)^\top + \sum_{x \in X_1} (x - \mu_1)(x - \mu_1)^\top.$$

The *between-class scatter matrix* can be defined as

$$S_B = (\mu_1 - \mu_0)(\mu_1 - \mu_0)^\top,$$

up to a constant scaling. For two classes, S_B has rank 1.

5.2 Optimization formulation

LDA seeks a projection vector $w \in \mathbb{R}^d$ that maximizes the Rayleigh quotient

$$J(w) = \frac{w^\top S_B w}{w^\top S_W w}.$$

Intuitively, we want the projected class means to be far apart, while the within-class spread in the projected space is small.

Proposition 2. *Assume S_W is invertible. Then the maximizer of $J(w)$ (up to scaling) is*

$$w \propto S_W^{-1}(\mu_1 - \mu_0).$$

Idea of the proof. Write

$$J(w) = \frac{(w^\top (\mu_1 - \mu_0))^2}{w^\top S_W w}.$$

Let $v = S_W^{1/2} w$, where $S_W^{1/2}$ is a matrix such that $S_W^{1/2} (S_W^{1/2})^\top = S_W$. Then

$$J(w) = \frac{(v^\top S_W^{-1/2} (\mu_1 - \mu_0))^2}{v^\top v}.$$

For fixed $S_W^{-1/2}(\mu_1 - \mu_0)$, this quotient is maximized when v is parallel to $S_W^{-1/2}(\mu_1 - \mu_0)$, by the Cauchy–Schwarz inequality. Therefore,

$$w \propto S_W^{-1}(\mu_1 - \mu_0).$$

□

Thus in the two-class case, we do not need to solve a general eigenvalue problem; it suffices to solve a linear system $S_W w = \mu_1 - \mu_0$, possibly with a small regularization term to handle singular S_W .

5.3 Classification rule in 1D

Once we have w , we project a feature vector x to a scalar

$$z = w^\top x.$$

Let

$$m_0 = \frac{1}{n_0} \sum_{x \in X_0} w^\top x, \quad m_1 = \frac{1}{n_1} \sum_{x \in X_1} w^\top x$$

be the projected class means. A natural decision threshold is the midpoint

$$\tau = \frac{m_0 + m_1}{2}.$$

Then the LDA classifier is

$$\hat{y}(x) = \begin{cases} 1, & \text{if } w^\top x \geq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

6 Numerical conditioning and the matrix $A^\top A$

6.1 Condition number

The (2-norm) condition number of an invertible matrix A is

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2.$$

For a symmetric positive definite matrix, $\|A\|_2$ is its largest eigenvalue and $\|A^{-1}\|_2$ is $1/\lambda_{\min}(A)$, so

$$\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}.$$

6.2 Why $A^\top A$ can be problematic

Let $A \in \mathbb{R}^{m \times n}$ have singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. Then

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_r}, \quad \kappa_2(A^\top A) = \frac{\sigma_1^2}{\sigma_r^2} = \kappa_2(A)^2.$$

Thus forming $A^\top A$ *squares the condition number*, potentially magnifying numerical errors.

For example, when computing an SVD via the eigenvalue decomposition of $A^\top A$, this squaring of the condition number can reduce accuracy compared to algorithms that operate directly on A (such as bidiagonalization followed by symmetric QR).

6.3 Regularization in LDA

In the LDA setting, S_W may be singular or ill-conditioned, especially in high-dimensional problems with relatively few samples. A common remedy is to add a small multiple of the identity:

$$S_W^{(\text{reg})} = S_W + \lambda I_d,$$

with $\lambda > 0$ small (e.g. $\lambda = 10^{-6}$). We then solve

$$S_W^{(\text{reg})} w = \mu_1 - \mu_0.$$

This improves conditioning and stabilizes the computation of w .

7 Summary

The project combines several ideas:

- The SVD factorization $A = U\Sigma V^T$ and its interpretation via singular values and singular vectors.
- Truncated SVD as the best rank- k approximation in Frobenius norm, motivating image compression.
- The relationship between the singular values of A and the eigenvalues of $A^T A$.
- The power method as a basic iterative algorithm for computing dominant eigenpairs of symmetric matrices.
- PCA as an eigenvalue/SVD-based method for unsupervised dimensionality reduction.
- Two-class LDA as a supervised method that uses scatter matrices and an $S_W^{-1}(\mu_1 - \mu_0)$ direction for classification.
- Numerical conditioning issues arising from forming $A^T A$, and the use of regularization in LDA.

These tools provide the mathematical foundation for the implementation tasks: computing SVD-based features from images, designing rank- k approximations, and training an LDA classifier in a low-dimensional feature space.